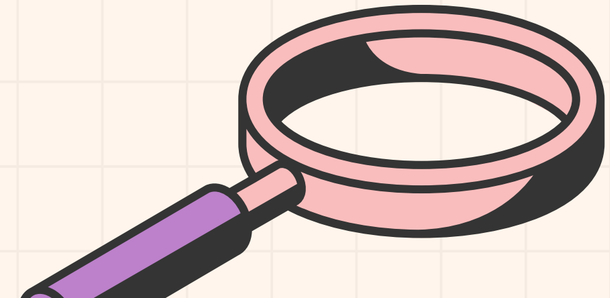
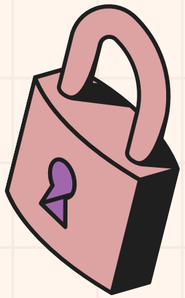
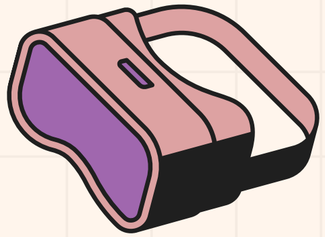
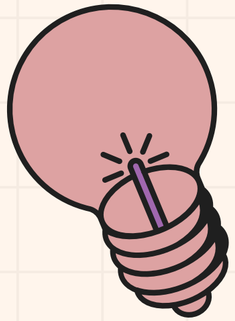


# AUTONOMOUS MANUFACTURING SYSTEM FOR SEMICONDUCTOR CHIP PRODUCTION





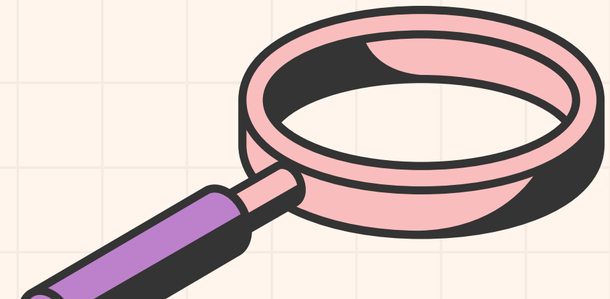


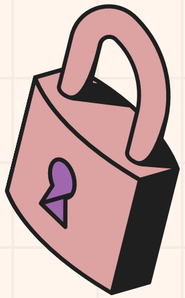


KELOMPOK :

AZAM AZRI AHMAD - 24/552221/PTK/16448

LAZUARDI IMAM SANTOSA - 24/550879/PTK/16359





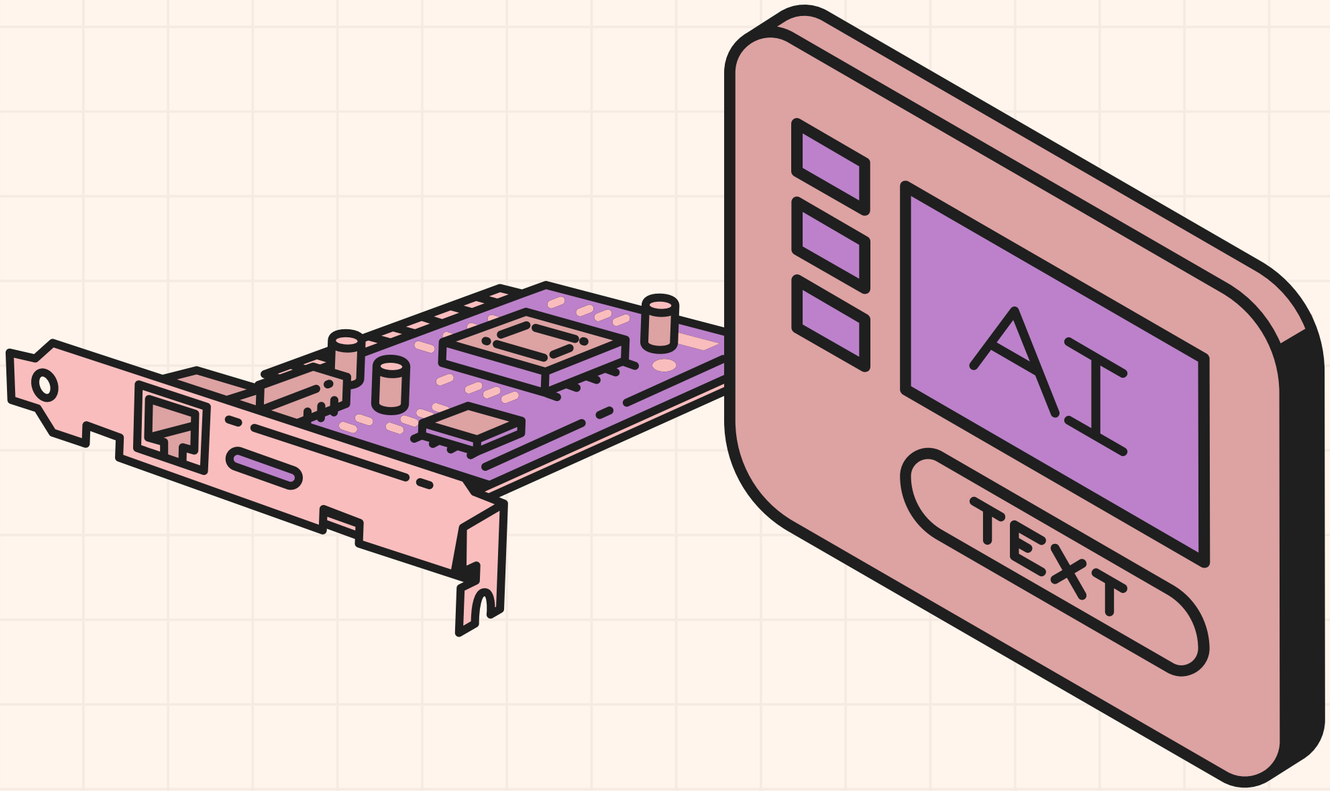
# INTRODUCTION

Semiconductor fabrication is the process of manufacturing electronic components (such as chips, transistors, and ICs) from semiconductor materials, primarily silicon. This process involves a series of complex stages, including:

- Design – The development of integrated circuit layouts using specialized software.
- Wafer Fabrication – Cutting pure silicon into thin slices called wafers.
- Photolithography – Printing circuit patterns onto wafers using light and masks.
- Doping & Etching – Modifying the electrical properties of the wafer and etching specific layers.
- Plating & Interconnection – Adding metal layers to connect the various components.
- Testing & Packaging – Performing quality checks and assembling the chips into protective casings.

This process is carried out in advanced facilities (called fabs) with ultra-clean environments to prevent contamination. The result is a semiconductor chip that serves as the "brain" of modern electronic devices, such as smartphones, computers, and autonomous vehicles.

Advancements in semiconductor fabrication (e.g., 5nm, 3nm technologies) continue to drive innovation in computing, artificial intelligence (AI), and the Internet of Things (IoT).




**sumber: id.wikipedia.org/wiki/Fabrikasi_semikonduktor**

# **PERUSAHAAN** FABRIKASI SEMI KONDUKTOR

### SAMSUNG

Samsung Semiconductor (South Korea)
Produces:

- Memory: DRAM and NAND Flash (used in smartphones and data centers)
- Logic Chips: Exynos (mobile processors) and AI chips such as HBM3

Technologies:

- 3D packaging to improve thermal efficiency
- GAAFET (Gate-All-Around) lithography for advanced 2nm chip production

### TSMC

- Produces:
- Advanced chips for fabless companies such as Apple (A-series/M-series), Nvidia (AI GPUs), AMD (Ryzen/EPYC processors), and Qualcomm (Snapdragon SoCs).
- Technologies:
- Leading the production of 3nm and 2nm chips using EUV (Extreme Ultraviolet) lithography.
- Holds approximately 60% of the global foundry market share.

### INTEL

Intel (USA)
Produces:

- CPUs: Core i-series for consumer devices and Xeon processors for data centers.
- GPUs: Arc series for gaming and AI applications.
- Foundry Services: Custom chips through Intel Foundry Services (IFS) for external clients.

Technologies:

- Intel 18A process (1.8nm) featuring RibbonFET transistor architecture, which enhances performance and power efficiency.
- Strategic focus on AI chips, including the Gaudi 3 accelerator for machine learning and deep learning workloads.

**sumber: www.tsmc.com/english , semiconductor.samsung.com, www.intel.co.id**

# **OTOMATIASI PADA** FABRIKASI SEMIKONDUKTOR

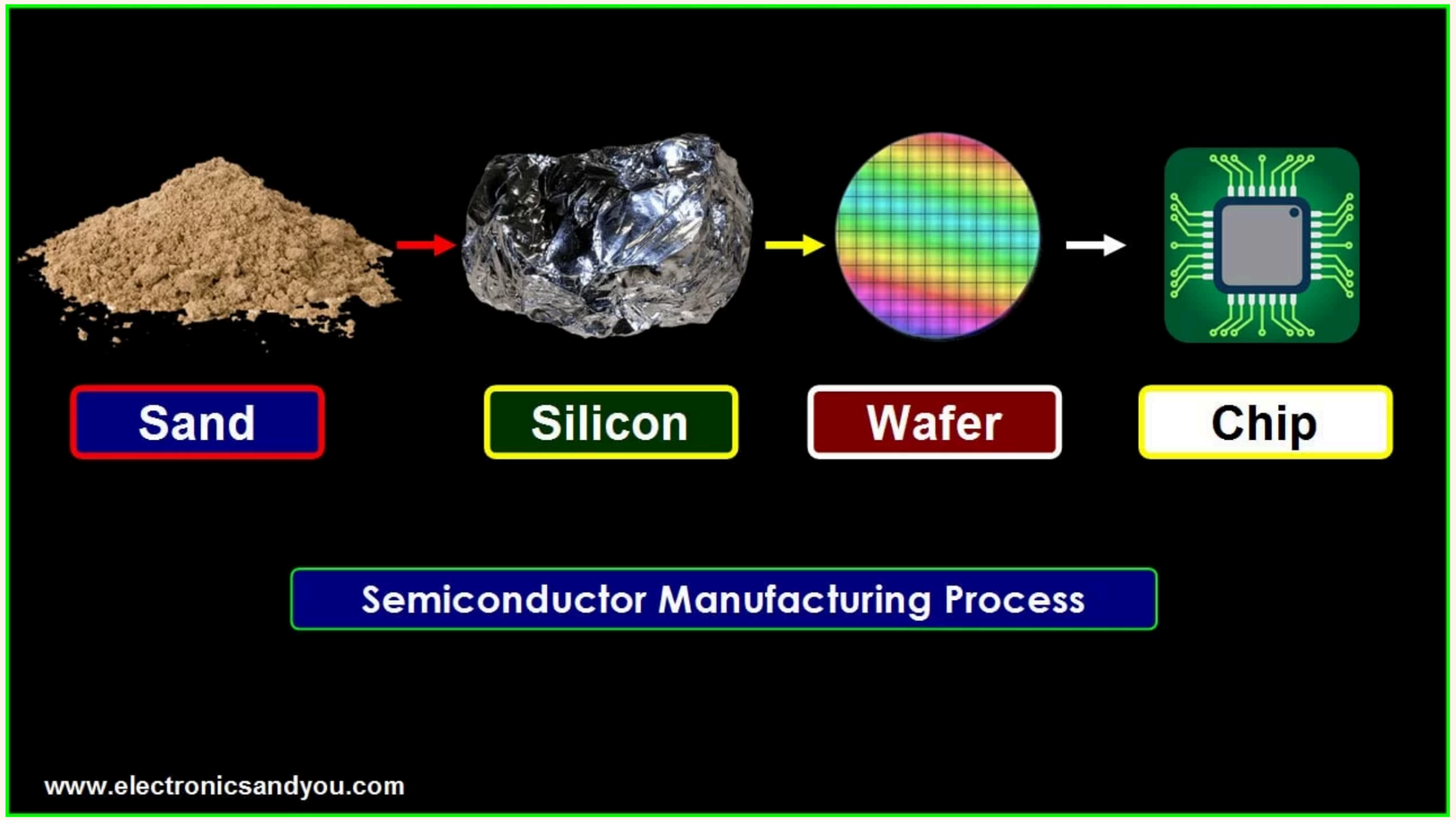
Automation in semiconductor fabrication plays a crucial role in enhancing efficiency, quality, and consistency in the manufacturing process. With the support of technologies such as artificial intelligence (AI) and machine learning (ML), production processes have become more advanced through predictive capabilities—such as detecting potential equipment failures early, allowing preventive actions to be taken before significant losses occur.

Systems like Manufacturing Execution Systems (MES) and Automated Test Equipment (ATE) enable real-time monitoring and automated chip testing, ensuring consistently high product quality. Additionally, the use of robotics for handling wafers and other sensitive materials helps minimize human error and improves the precision of manufacturing operations.

Automation in semiconductor fabrication comes in three types:

1. Fully Automated (where all processes are carried out automatically).
2. Semi-Automated (where some processes are handled by humans, such as moving wafers to different stages of production).
3. Manual (where most or all processes are performed by humans).

# SEMICONDUCTOR FABRICATION




This image illustrates the semiconductor manufacturing process in four main steps:

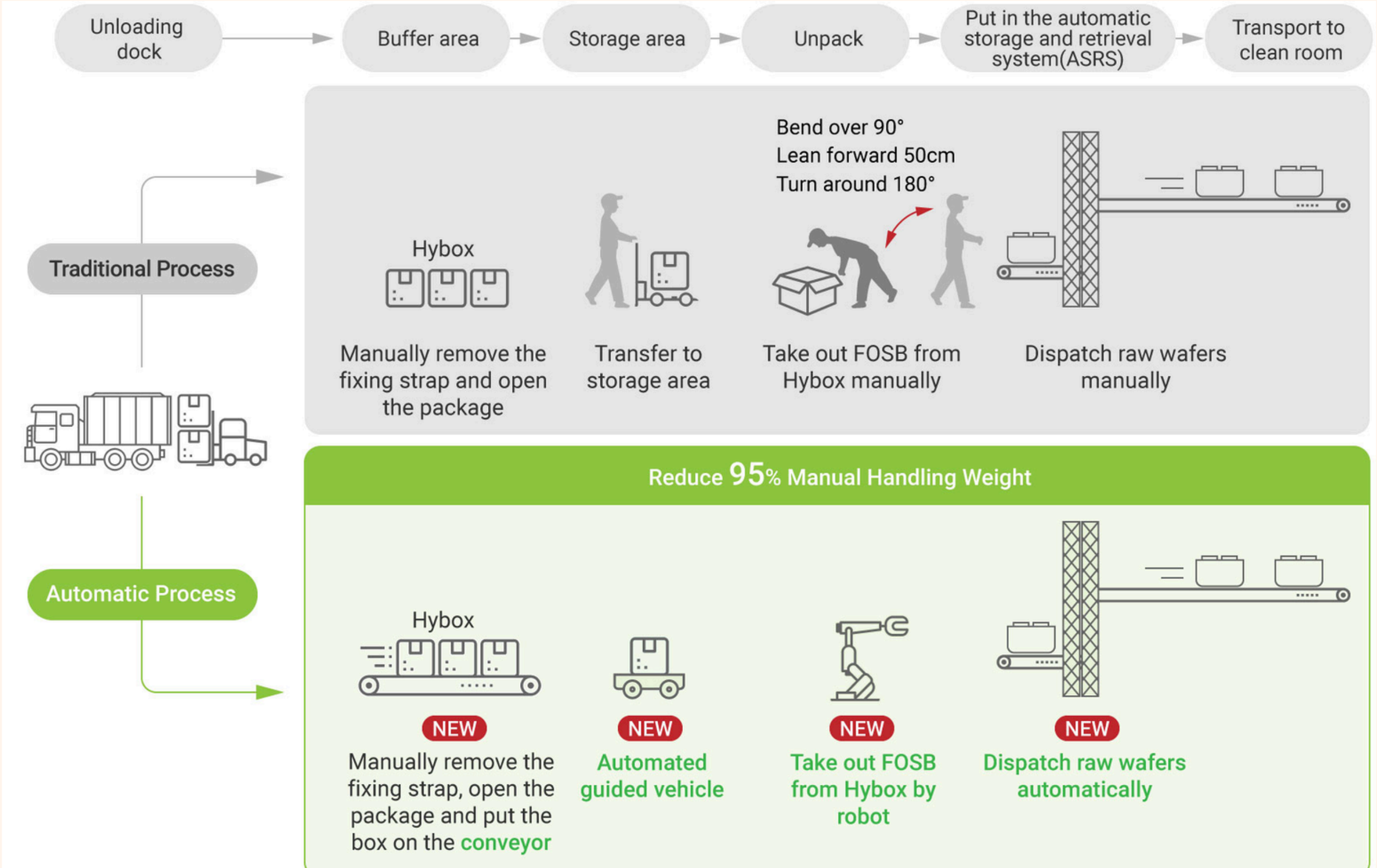
1. Sand – The raw material, rich in silicon.
2. Silicon – Extracted and purified from sand.
3. Wafer – Sliced and polished silicon formed into wafers.
4. Chip – Electronic circuits are built on the wafer to create chips.

It shows how common sand is transformed into advanced microchips.

# AUTOMATION IN SEMICONDUCTOR FABRICATION

Receiving Area

**Automated Handling System for Wafer Warehouses**



# AUTOMATION IN SEMICONDUCTOR FABRICATION

→ Cleanroom
**Automated Wafer Inbound/Outbound System**
NEW

| Steps | Receive | Unpack | Put-away | Pack | Ship Out |
|---|---|---|---|---|---|
| Traditional Process | Manual Receive | Manual Unpack | Manual Put-away / Pick-up | Manual Pack | Manual Transfer |
| Automatic Process | Auto-retrieval Conveyor | 8 Steps of Auto-unpack | OHT Linkage for Stocker In/Out Transfer | 8 Steps of Auto-pack | Auto-ship out Conveyor |

Reduced 13 Million Manual Tasks per Year

Enhanced Efficiency by 70%











**Unpacking**

Pick → Turn → Cut → Stretch → Takeoff → Throwaway → Pluck → Out

**Packing**

Pick → Label → Stretch → Desiccate → Pack → Seal → Fold → Glue

# AUTOMATION IN SEMICONDUCTOR FABRICATION

Finished Goods Warehouse
**Automated Wafer Packing and Sorting System**
NEW

| Steps | Receive | Storage | Pick | Pack | Sort/Pick-up |
|---|---|---|---|---|---|
| Traditional Process | Manual Receive | Storage | Manual Pick | Manual Pack | Manual Sort and Search |
| Automatic Process | Cleanroom/ FG Warehouse Auto-linkage | ASRS | Auto-retrieval Based on Requirement | Auto-pack | Auto-sort |

**314 Tons of Monthly Workload Reduction for Warehouse Staff**

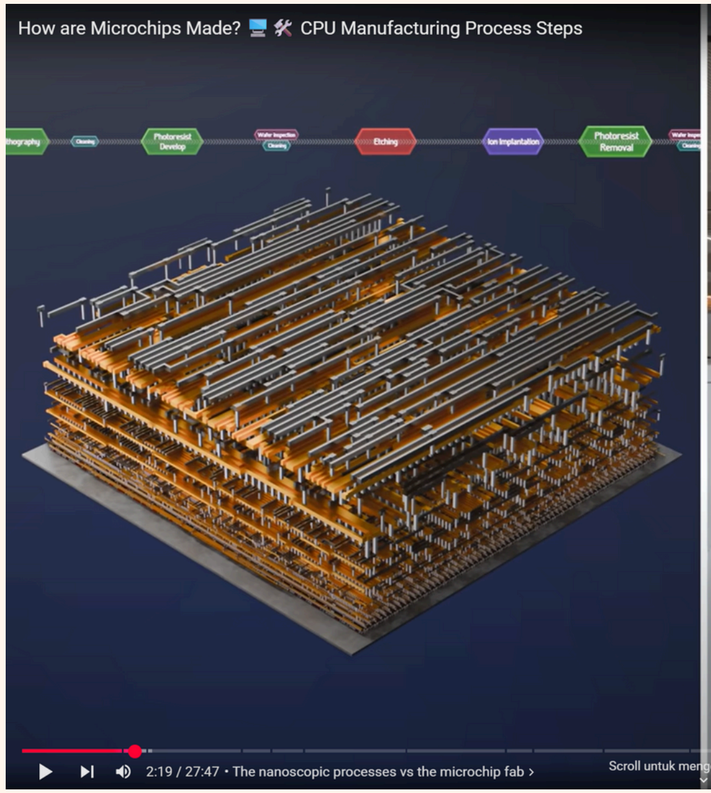
**35% Improvement in Efficiency for Wafer Delivery Drivers**





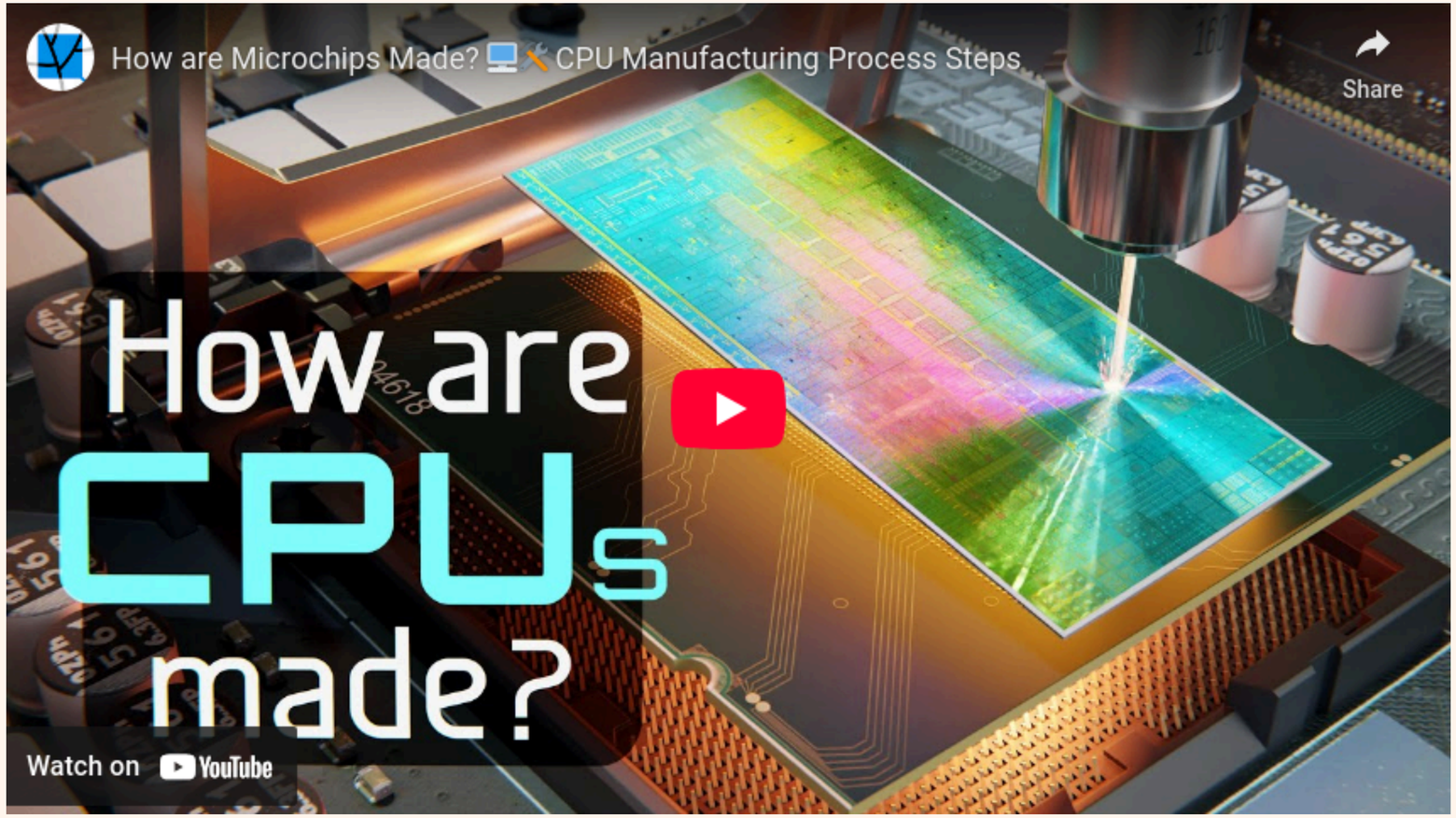
# AUTOMATION IN SEMICONDUCTOR FABRICATION



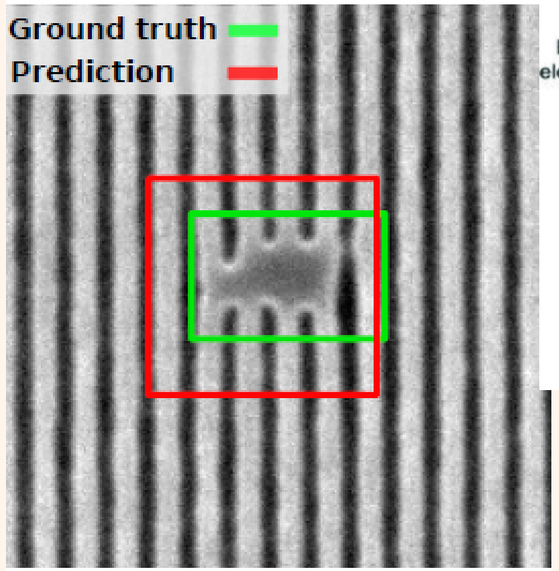


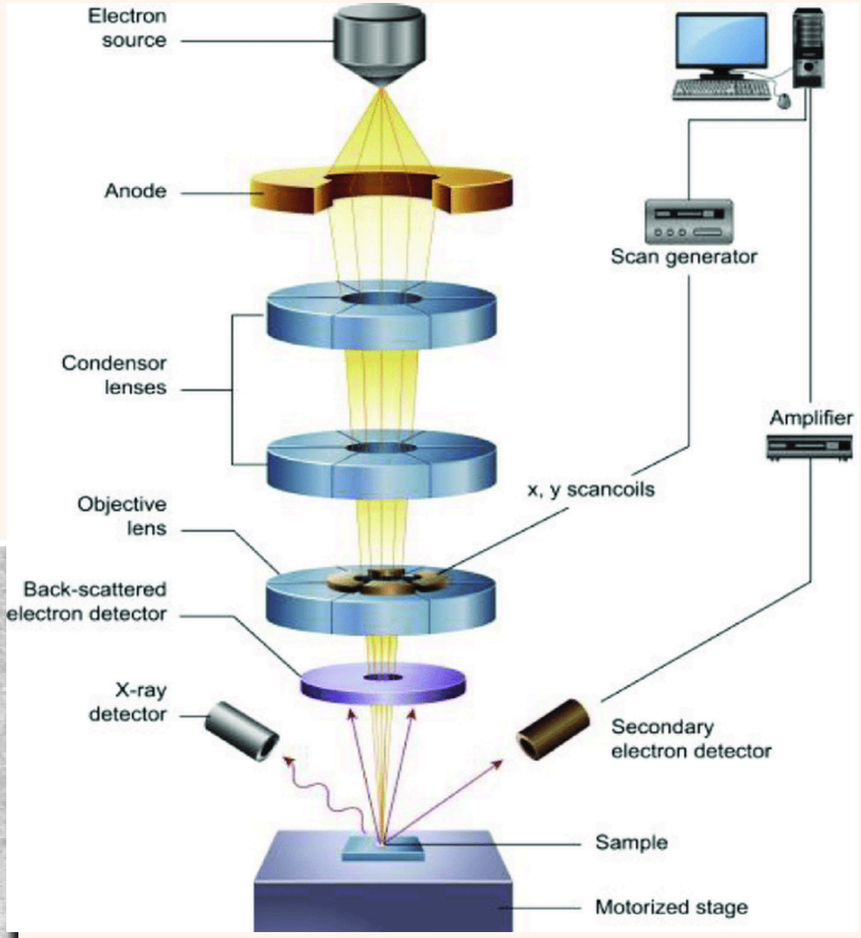

# AUTOMATION IN SEMICONDUCTOR FABRICATION



# AUTOMATION IN SEMICONDUCTOR FABRICATION

This study utilizes imaging systems based on Scanning Electron Microscopes (SEM), rather than conventional optical cameras, because SEM can produce ultra-high-resolution images at the nanometer scale—essential for detecting extremely small defects in modern semiconductor chips. Optical cameras are not capable of detecting defects smaller than 10 nm, making SEM the primary tool for inspection. The types of SEM used in this research include CD-SEM (Critical Dimension SEM), SEMVision™ and SEMVision™ cX from Applied Materials, as well as Hitachi Review SEM models such as the RS-5500EQEQ and RS4000. These instruments not only generate SEM images but also integrate automated defect classification and detection systems. The resulting images are processed using machine learning and deep learning algorithms, particularly convolutional neural networks (CNNs), to accurately and efficiently detect and classify defects within semiconductor manufacturing processes.



# CPS IN SEMICONDUCTOR FABRICATION

Definition of CPS:
Cyber-Physical System is a system that integrates computing (cyber), communication networks, and physical components (physics). In CPS, sensors, actuators, and control systems work together through networks to manage physical processes automatically and intelligently.

Semiconductor Fabrication as a CPS Implementation:
A semiconductor fabrication plant (fab) is a concrete example of Cyber-Physical Systems (CPS) implementation because:

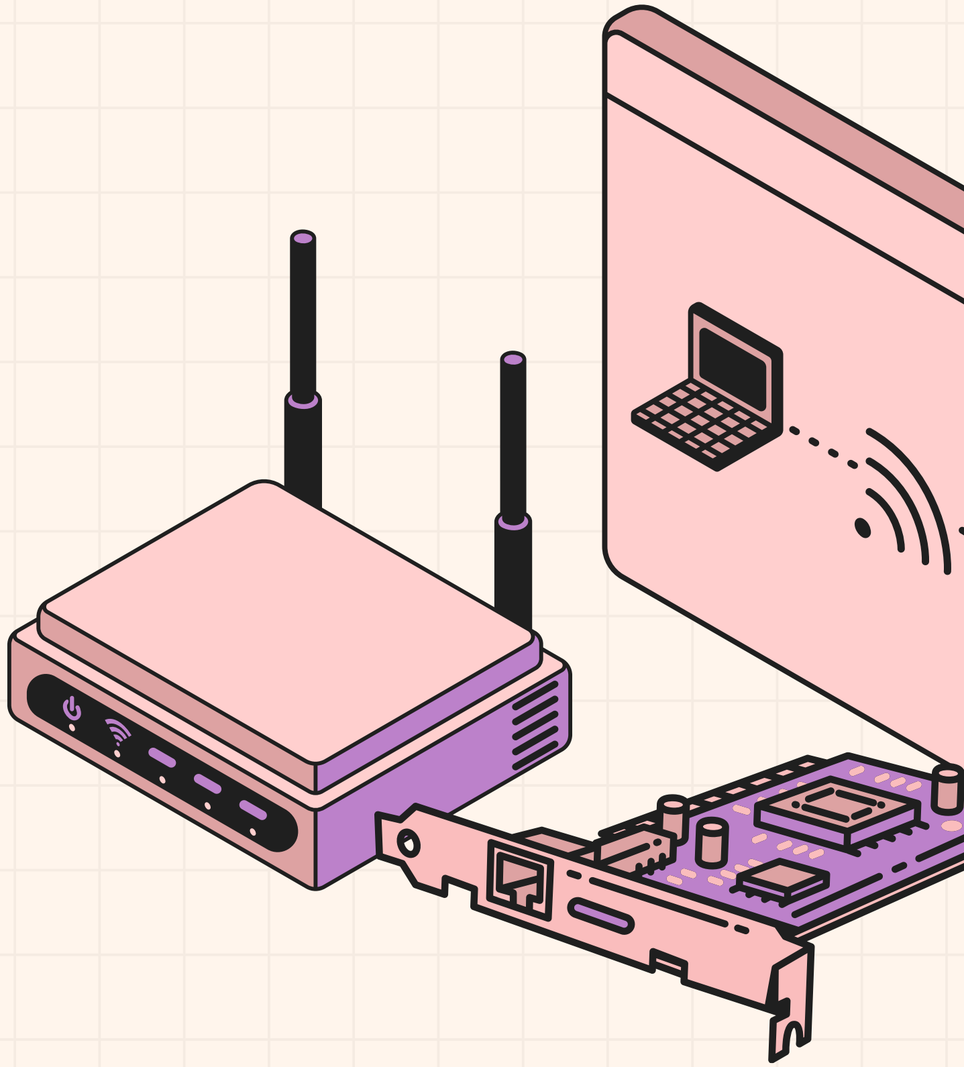
- It uses advanced sensors to monitor temperature, pressure, air cleanliness, and wafer quality.
- Data from the sensors is processed by computers or AI systems for automated decision-making (e.g., adjusting cleanroom temperature or shutting down machines experiencing anomalies).
- All systems and machines are interconnected through an industrial network (Industrial IoT) to operate in a synchronized manner.

# TYPE OF CPS

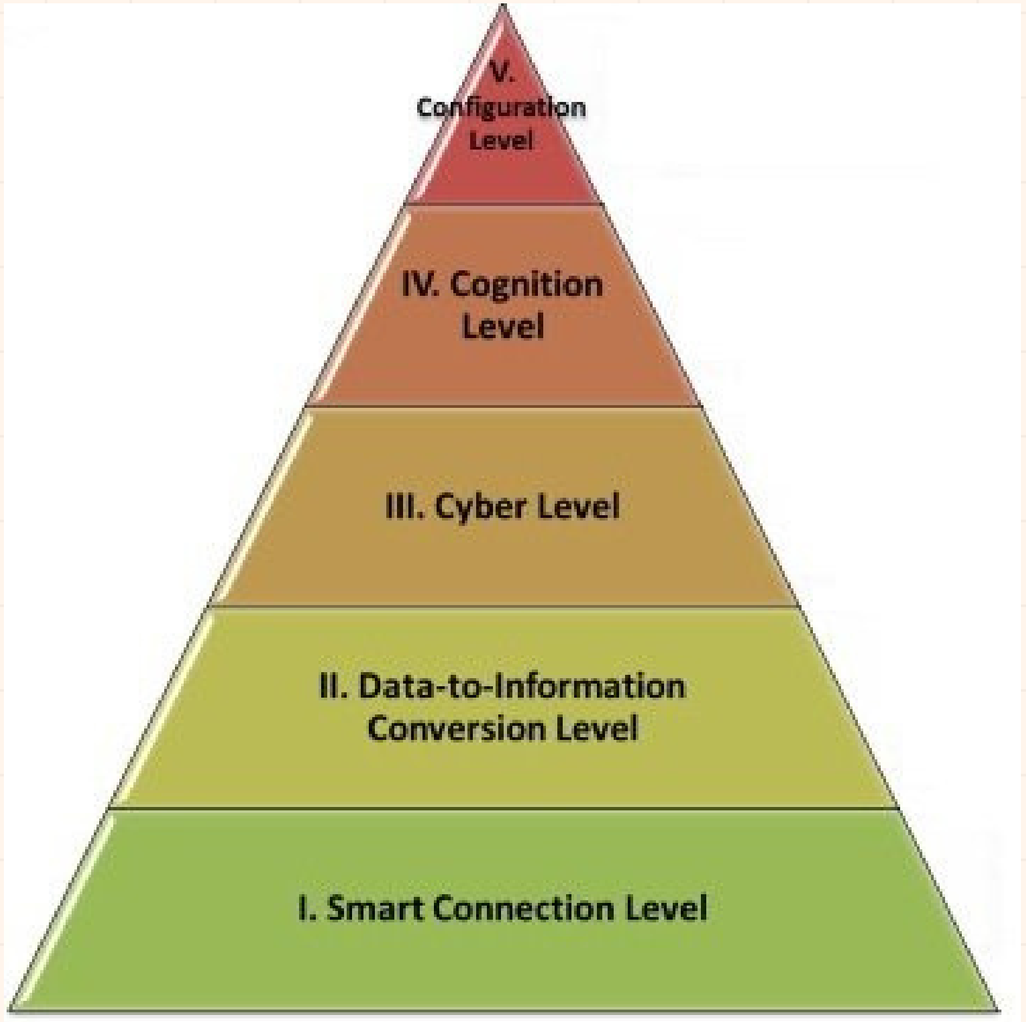
## AUTONOMOUS CYBER-PHYSICAL SYSTEM

A-CPS is a system that can operate independently, make its own decisions, and adapt to environmental conditions without continuous human intervention.

- The chip manufacturing process involves thousands of precision steps that are highly sensitive to human error.
- The system uses automated robotics, AI and real-time monitoring to control every step, from wafer handling to final testing.
- In some cases, the system also has cognitive elements, such as machine learning, to detect process anomalies and make predictive improvements.



# CPS ARCHITECTURE IN SEMICONDUCTOR FABRICATION




Control systems execute physical adjustments, such as halting a production line or fine-tuning lithography focus

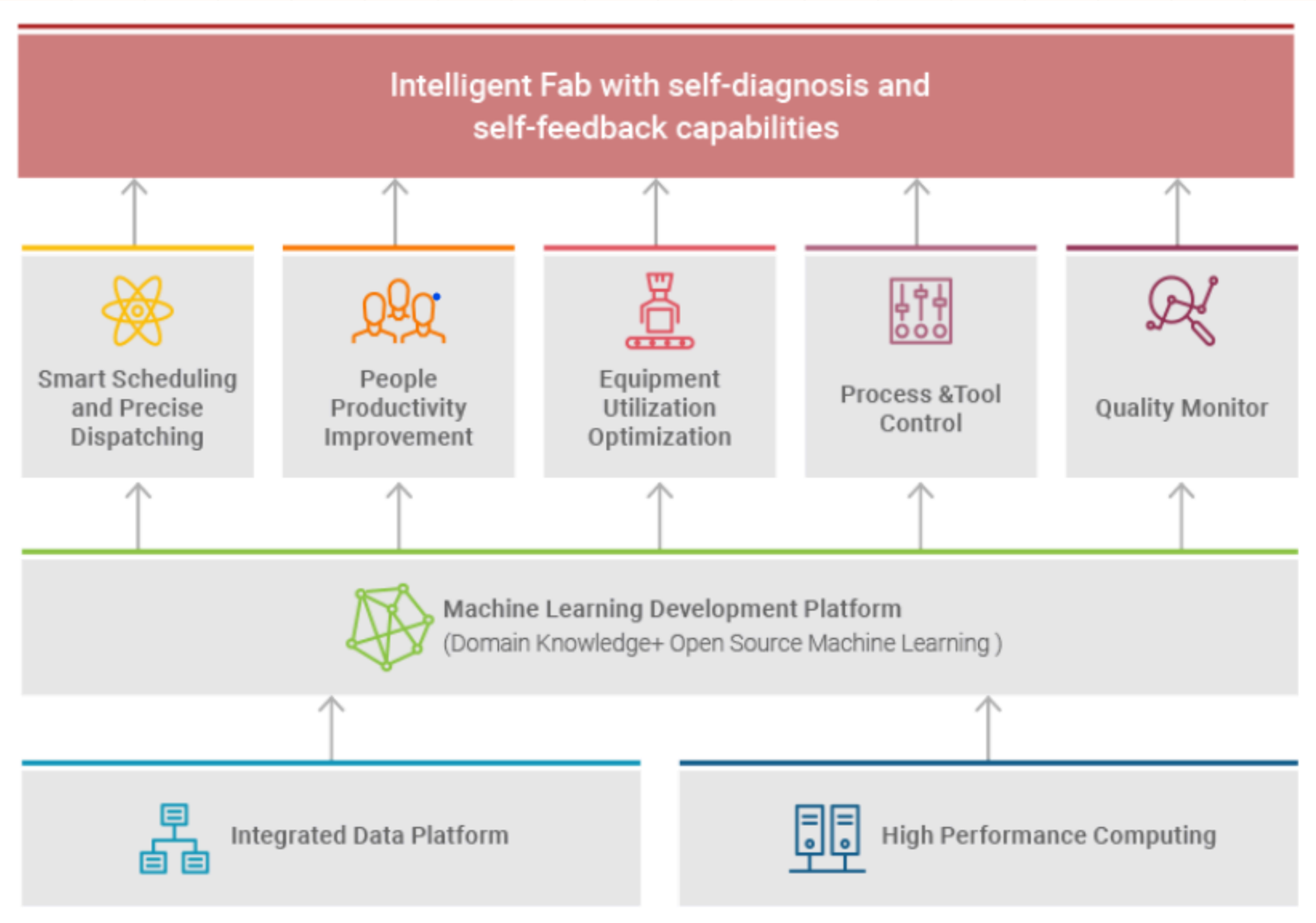
Automated decision-making systems evaluate outcomes and send commands to actuators.
Example: Auto-recalibration of etching tools when pattern drift is detected

AI and ML models process input data to detect anomalies and predict failures.
Tools: CNNs for wafer defect classification

Data preprocessing is performed at the edge, using embedded processors near the machines.
Example: Filtering sensor noise and extracting relevant parameters

Sensors monitor cleanroom conditions (temperature, humidity, particle count) and machine status.
Example: SEM (Scanning Electron Microscope) sensors with nanometer-scale resolution

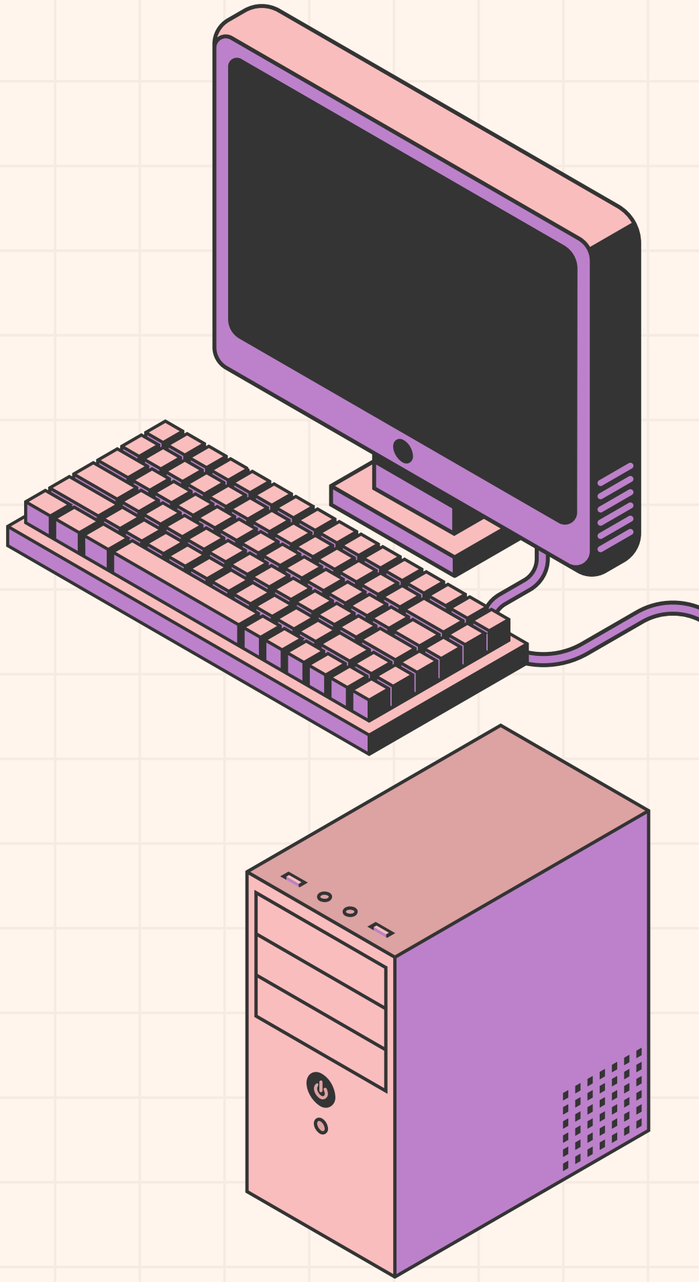
# IMPLEMENTATION : TSMC INTELLIGENT FABRICATION



# OBJECTIVES AND BENEFITS

## AUTONOMOUS CPS IN PRODUCTION CHIP

| Objective | Explanation | Benefit Areas | Explanation |
|---|---|---|---|
| Improve operational efficiency | The production process becomes faster and has minimal errors. | Production | Faster process, minimal human error. |
| Ensuring high product precision and quality | Processes such as photolithography and etching are very sensitive, so they require automated control. | Quality | Automated testing ensures that the chip results meet high standards. |
| Improve system security and stability | Reduce dependence on humans and prevent work accidents or major losses | Cost efficiency | More optimal use of energy and materials. |
| Integrate physical and logical systems comprehensively | All sensor data, processing and actions are combined in one integrated system. | Data analytics | Data & AI-based monitoring enables early problem prediction. |



# REFERENSI


Griffor, E.R. (NIST) (2017)
Framework for Cyber-Physical Systems: Volume 1 – Overview
https://doi.org/10.6028/NIST.SP.1500-201

https://www.tsmc.com/

Lee, E.A., & Seshia, S.A. (2017)
Introduction to Embedded Systems: A Cyber-Physical Systems Approach
http://leeseshia.org/

https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=66

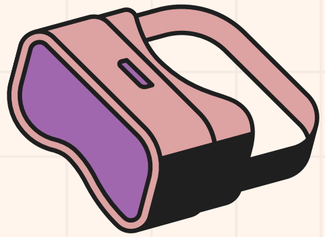
https://uk.marketscreener.com/quote/stock/TSMC-TAIWAN-SEMICONDUCTOR-6492349/news/Taiwan-Semiconductor-Manufacturing-TSMC-Pioneers-the-World-s-First-Automated-Wafer-Inbound-Outboun-37667073

# REFERENSI

Automated Semiconductor Defect Inspection in
Scanning Electron Microscope Images:
a Systematic Review
https://doi.org/10.48550/arXiv.2308.08376

https://esg.tsmc.com/en-US/articles/145

https://esg.tsmc.com/en/update/innovationAndService/caseStudy/41/index.html



# THANK YOU

